Euler's method

Now that we know that a direction field is a geometric view of a differential equation, and the solution curve through the direction field is the geometric view of its solution, we understand better what we're actually doing when we use an analytical method to find the solution to a differential equation.

Essentially, the general solution gives an equation that models all curves that can be sketched through the direction field, at least on a particular interval of validity. And when we solve an initial value problem, we solve for the single solution curve that passes through one specific point.

Numerical methods

But now we want to look at a third method for finding the solution to the differential equation. First, we looked at analytical methods for solving some special types of first and second order differential equations. Then we looked at the geometric method of sketching the direction field and using it as a guide to sketch solution curves. Now we want to look at using a numerical method to approximate the solution.

Numerical methods are particularly useful when we aren't able to use an analytic approach to find the equation of the solution. In fact, in the real world, most differential equations are solved with numerical methods, because most differential equations don't follow the perfectly near format of a linear equation, homogeneous equation, exact equation, etc.

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The numerical method we'll look at here is called Euler's method, and as long as we have the differential equation and one initial condition, we can use it to approximate the solution.

Euler's method is an iterative process in which we repeat the same step over and over. The more steps we use, the better approximation we'll get. We choose a **step size** Δt, which is the distance between two successive values of t at which y is calculated.

Then we use the initial condition to identify (t_0, y_0), and we plug (t_0, y_0) and Δt into Euler's formula,

$$y_1 = y_0 + [f(t_0, y_0)]\Delta t$$

in order to calculate y_1. The value of t_1 is $t_1 = t_0 + \Delta t$. Then we can use (t_1, y_1) and Δt to calculate y_2, use (t_2, y_2) and Δt to calculate y_3, etc. We continue this calculation again and again, until we arrive at the value of t at which we wanted to approximate the value of y.

When we use Euler's method to approximate the solution at a particular point, it's helpful to build a table to keep all of our calculations organized.

t	y	**Result**
t_0	y_0	y_0
t_1	$y_1 = y_0 + [f(t_0, y_0)]\Delta t$	y_1
t_2	$y_2 = y_1 + [f(t_1, y_1)]\Delta t$	y_2
...	...	...

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t_n $y_n = y_{n-1} + [f(t_{n-1}, y_{n-1})]\Delta t$ y_n

With all that in mind, let's work through an example so that we can see this method in action.

Example

Use Euler's method and two steps to find $y(1)$, given $y(0) = 2$.

$$y' = y - t$$

We're starting with the initial condition $y(0) = 2$ and attempting to approximate $y(1)$, which means we're starting at $t_0 = 0$ and ending at $t = 1$. To find step size, we divide the difference between these by the number of steps we were asked to use, in our case, by 2.

$$\frac{1-0}{2} = \frac{1}{2} = 0.5 = \Delta t$$

Now we can start building our table. The values of t_0 and y_0 both come from our initial condition, $y(0) = 2$. To find t_1 and t_2, we add Δt to the previous value for t. We do that until we reach $t_n = 1$, which is the value the problem originally asked for when it asked us to find $y(1)$.

$t_0 = 0$ $y_0 = 2$ $y_0 = 2$

$t_1 = 0 + 0.5 = 0.5$

$t_2 = 0.5 + 0.5 = 1$

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Now we'll use Euler's formula with (t_0, y_0) and Δt to find y_1.

$$t_0 = 0 \qquad y_0 = 2 \qquad\qquad y_0 = 2$$

$$t_1 = 0.5 \qquad y_1 = 2 + (2 - 0)(0.5) \qquad\qquad y_1 = 3$$

$$t_2 = 1$$

Finally, we'll use Euler's formula with (t_1, y_1) to find y_2, which is the value corresponding to $t_2 = 1$, which is the value we were asked to approximate.

$$t_0 = 0 \qquad y_0 = 2 \qquad\qquad y_0 = 2$$

$$t_1 = 0.5 \qquad y_1 = 2 + (2 - 0)(0.5) \qquad\qquad y_1 = 3$$

$$t_2 = 1 \qquad y_2 = 3 + (3 - 0.5)(0.5) \qquad\qquad y_2 = 4.25$$

So Euler's method and $y(0) = 2$ with two steps approximated $y(1)$ as

$$y(1) \approx 4.25$$

We'll look at one more example of a problem worded slightly differently. If we're working with decimal numbers, one thing we want to remember with Euler's method approximations is to keep as many decimal places as possible as we work through the approximation.

If we start rounding values from earlier steps, our approximation will get less and less accurate as we go. So it's best to keep more decimal places as we work through each step, and then only round the final answer.

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Example

Use Euler's Method with step size 0.2 to approximate the values y_1, y_2, y_3, y_4, and y_5, when $y(0) = 5$.

$$y' = y - 3t + t^2$$

Since we're given step-size directly, we already know that

$$\Delta t = 0.2$$

To start building our table, we identify $t_0 = 0$ and $y_0 = 5$ from our initial condition, $y(0) = 5$. Since we need to find y_5, we'll start adding $\Delta t = 0.2$ to t_0 until we reach t_5.

$t_0 = 0$ $\quad$ $y_0 = 5$ $\quad$ $y_0 = 5$

$t_1 = 0.2$

$t_2 = 0.4$

$t_3 = 0.6$

$t_4 = 0.8$

$t_5 = 1$

Now that we've built the outline for our table, we'll use Euler's formula to find y_1, y_2, y_3, y_4, and y_5.

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$t_0 = 0$	$y_0 = 5$	$y_0 = 5$
$t_1 = 0.2$	$y_1 = 5 + (5 - 3(0) + 0^2)(0.2)$	$y_1 = 6$
$t_2 = 0.4$	$y_2 = 6 + (6 - 3(0.2) + 0.2^2)(0.2)$	$y_2 = 7.088$
$t_3 = 0.6$	$y_3 = 7.088 + (7.088 - 3(0.4) + 0.4^2)(0.2)$	$y_3 = 8.2976$
$t_4 = 0.8$	$y_4 = 8.2976 + (8.2976 - 3(0.6) + 0.6^2)(0.2)$	$y_4 = 9.66912$
$t_5 = 1$	$y_5 = 9.66912 + (9.66912 - 3(0.8) + 0.8^2)(0.2)$	$y_5 = 11.250944$

After filling out the table, we can say that the value of $y(1)$ is approximately

$$y(1) \approx 11.250944$$

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